

DRUG	ROUTE	PAEDIATRIC DOSE	ADULT DOSE	REMARKS
ANTIBIOTICS				
*For neonates ≤ 28 days, please refer to "Neonatal Antibiotics" or check Neofax for appropriate dosing.				
Amikacin	IV	15mg/kg 24H	1g	Conc ≤ 5mg/mL, over ≥30 min. If obese*, use dosing weight of IBW + 0.4(TBW – IBW) for aminoglycosides Pre-4th trough if >3 days, aim <2.3
Amoxicillin	Oral	15-30 mg/kg TDS	1g	Acute OM/severe pneumonia: dose 90mg/kg/DAY 20-45mg/kg BD for compliance Strep throat: 25mg/kg BD (Max 500mg/dose)
Amoxicillin/ clavulanic acid (Augmentin)	Oral	15-25mg Amox-clav /kg BD Severe: 22.5mg Amox-clav /kg TDS	Susp: 12.5mL Tab: 625mg	Dose expressed in amox/clav Max clavulanic dose 10mg/kg/day (125mg/dose) Max suspension dose 12.5mL (= amoxicillin component 500mg) Susp: 228mg/5ml (200mg/5ml amox) Tab: 625mg (500mg amox)
	IV	30mg Amox-clav /kg 8H	1.2g	
Ampicillin	IV	Neonate (≤28 days): SEE PAGE 2 Mild-mod infections: 25mg/kg 6H Sepsis: 50mg/kg 6H Meningitis: 75-100mg/kg 6H	500mg 1 g 2 g	Recommended conc ≤ 30mg/mL
Ampi-Sulbactam (Unasyn)	Oral	<30kg: 12.5-25mg Unasyn /kg BD ≥30kg: 375-750mg Unasyn BD	750mg	Dose expressed in Ampi/sulbactam Tab: 375mg
	IV	Mild-mod: 75mg Unasyn /kg 6H Severe: 150mg Unasyn /kg 6H	3g	1 vial: 1000mg ampicillin, 500mg sulbactam
Azithromycin	Oral/IV	10mg/kg 24H for 3 days Alternative: 10mg/kg on Day 1, then 5mg/kg on Days 2-5	500mg	Susp: 200mg/5mL, Tab: 250mg
Cefaclor	Oral	10-15mg/kg TDS	500mg	Potential for serum sickness Susp: 125mg/5mL
Cefazolin	IV	20mg/kg 8H Severe: 30-50mg/kg 8H	2g	For septic arthritis: 50mg/kg 8H
Cefepime	IV	50mg/kg 12H Severe: 50mg/kg 8H	1-2g	Use 8H dosing for neutropenic fever in paediatric oncology patients
Cefotaxime	IV	50mg/kg/dose 8H Meningitis: 75-100mg/kg/dose 8H	2g	
Ceftazidime	IV	30mg/kg 8H Severe: 50mg/kg 8H	1-2g	Use 50mg/kg dosing for neutropenic fever in paediatric oncology pts
Ceftriaxone	IV	Sepsis: 50mg/kg 24H Meningitis: 50mg/kg 12H (100mg/kg 24H if <20kg) Typhoid Fever: 80-100mg/kg 24H	2g	If total daily dose for meningitis or typhoid exceeds 2g/dose, split total dose into 12H. For OPAT purposes only: total daily dose can be given as single daily dosing not exceeding 4g/day
Cefuroxime	Oral	Suspension: 10-15mg/kg BD Tablets (>12yo): 250-500mg BD	500mg	Susp: 125mg/5mL, Tab 250mg (should not be crushed), suspension and tablet dosing are NOT interchangeable
Cephalexin	Oral	10-15mg/kg QDS Severe: 15-25mg/kg QDS	500mg	Susp: 125mg/5mL, Capsule: 250mg Split total daily dose to 3 divided doses for compliance
Clarithromycin	Oral	7.5mg/kg BD	500mg	Susp: 125mg/5mL, Tab 500mg
Clindamycin	Oral/IV	Mild: 5-7mg/kg/dose 8H Mod/severe: 7-13mg/kg/dose 8H	IV: 900mg PO: 450mg	May cause pseudomembranous colitis Capsule: 150mg
Cloxacillin	Oral	6.25 – 12.5mg/kg QDS	500mg	Susp: 125mg/5mL, Capsule: 250mg
	IV	25 – 50mg/kg 6H	1-2g	
Doxycycline	Oral	2.2mg/kg/dose 12H	100mg	Avoid in <8YO
Gentamicin	IV	≥28 days old: 6mg/kg 24H	360mg	If obese*, use dosing weight of IBW + 0.4 (TBW – IBW). Conc ≤ 10mg/mL Pre 3rd trough if >3 days - aim <1

* Definition of obesity: >97th centile for weight | Ideal weight can be derived using 50th centile weight (especially for aminoglycosides), or weight centile corresponding to height centile

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ANTIBIOTICS (Continued)				
<i>*For neonates ≤ 28 days, please refer to "Neonatal Antibiotics" or check Neofax for appropriate dosing.</i>				
Meropenem	IV	Mild: 10mg/kg 8H Severe: 20mg/kg 8H Meningitis: 40mg/kg 8H	1g 1g 2g	
Metronidazole	Oral/IV	10mg/kg 8H	500mg	Tablet = 200mg
Penicillin G	IV	0.05MU/kg 6H Severe/Meningitis: 0.1MU/kg 6H	2 – 4 MU	Order in million units instead of mg. 1 million units = 600mg
Penicillin V	Oral	7.5-15mg/kg QDS	500mg	Only available as 250mg tablet
Piperacillin/ Tazobactam	IV	60-90mg Pip-tazo /kg 6H	4.5g	Dose expressed in piperacillin/tazo . Infusion conc ≤90mg/mL over 30min
Trimethoprim/ sulfamethoxazole	Oral/IV	Treatment: 4-6mg TMP /kg 12H <i>S. Maltophilia</i> : 5mg TMP /kg 8H PCP treatment: 5mg TMP /kg 6-8H	160mg	Dose expressed in Trimethoprim (TMP) component Susp: 40mg/200mg (TMP/SMX) in 5ml Tab: 80mg TMP Refer to individual protocols for PCP prophylaxis regimens
Vancomycin	IV	<16 years: 15mg/kg 6H ≥16 years: 15mg/kg 12H	1g	Check pre-3rd - 4th dose trough Conc ≤5mg/mL over ≥60mins
	Oral	10mg/kg 6H	125mg	Suspension: 50mg/mL (prepared with IV vial) For <i>C. diff.</i> only – 1 st line for severe <i>C. diff.</i> may dose up to 500mg/dose
NEONATAL ANTIBIOTICS (POSTMENSTRUAL AGE ≥36 WEEKS AND ≤28 DAYS OF LIFE)				
<i>*Use BIRTH WEIGHT if admission weight is less than birth weight</i>				
<i>*For PMA <36 weeks, refer to Neofax doses</i>				
Amoxicillin	Oral	15mg/kg/dose 12H		
Ampicillin	IV	<u>For Group B Streptococcal meningitis</u> 0 to 7 days of age: 100mg/kg/dose 8H 8 days and older: 75mg/kg/dose 6H		For other indications: Refer to Neofax
Cefazolin	IV	25mg/kg/dose		For PMA 37 – 44 weeks PND 0 – 7 12H PND > 7 8H
Cefotaxime	IV	<u>To consider use in Meningitis:</u> 0 to 7 days of age: 50mg/kg/dose 8 – 12H 8 days or older: 50mg/kg/dose 6 – 8H		Consider longer intervals for neonates < 2kg
Gentamicin	IV/IM	4mg/kg/dose 24H		Do pre-3rd trough level if treatment is anticipated to be >5 days
Metronidazole	Oral/IV	Loading dose: 15mg/kg/dose Maintenance dose 7.5mg/kg/dose 6 – 8H		PMA 34 – 40wks 8H > 40wks 6H
Vancomycin	IV	10 – 15 mg/kg		For PMA 37 – 44 weeks PND 0 – 7 12H PND > 7 8H
ANTI-VIRALS				
Acyclovir	IV	<u>HSV encephalitis:</u> Term – 3mo: 20mg/kg 8H >3 mo – 12 years: 15mg/kg 8H >12 years 10mg/kg 8H <u>VZV/ Eczema herpeticum</u> 10mg/kg 8H		If BMI > 30kg/m ² or >97th weight percentile, use ideal body weight for dose calculation *Max conc for infusion ≤ 5mg/mL Ensure patient is well hydrated *Check renal function
	Oral	<u>VZV (within 24 hrs of rash):</u> 20mg/kg QDS for 5 days <u>HSV gingivostomatitis:</u> 20mg/kg/dose QDS	800mg	
Valacyclovir	Oral	<u>VZV:</u> 20mg/kg TDS for 5 days <u>HSV gingivostomatitis:</u> 20mg/kg BD for 5 to 7 days	1000 mg	Minimum age: 3 months

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ANTI-VIRALS (Continued)				
Oseltamivir (Tamiflu)	Oral	≤8mo: 3mg/kg BD 9mo - <12mo: 3.5mg/kg BD <u>≥12mo: Use absolute weight dosing</u> ≤15kg: 30mg BD >15-23kg: 45mg BD >23-40kg: 60mg BD >40kg: 75mg BD	75mg	BD for treatment; OD for prophylaxis Treatment duration: 5 days Only available as 75mg capsule *Dose adjust in renal-impairment
ANTI-FUNGALS				
Anidulafungin	IV	Day 1: 3mg/kg/dose (loading) Subsequent days: 1.5mg/kg OD	200mg 100mg	
Fluconazole	Oral/IV	Day 1: 6-12mg/kg OD Subsequent days: 3-12mg/kg OD	800mg	Dosing depends on indication IV: 15mmol Na per 100mL
Micafungin	IV	Treatment: 2-4mg/kg OD Prophylaxis: 1mg/kg OD	200mg 50mg	For treatment of oesophageal candidiasis, max dose 150mg .
ANTI-PYRETICS / ANALGESIA				
Paracetamol	Oral/IV	10-15mg/kg 6H	1g	Use higher end of dose range for analgesia. Max dose 75mg/kg/day *Caution in hepatic failure. Susp: 120mg/5ml Rectal: 125mg, 650mg
	Rectal	10-20mg/kg/dose 6H		
Ibuprofen	Oral	5-10mg/kg/dose 6-8H	200-400mg 1.2g/day	Use in children ≥ 6 months *Use lower range for anti-pyretic effect, higher range for anti-inflammatory effect. Max dose 40mg/kg/day *Caution in renal impairment, dehydration, asthma. Risk of bleeding
Diclofenac	Oral/Rectal	0.5-1mg/kg 8-12H	50mg	Use in children ≥ 1 yo Rectal: 12.5mg, 50mg *Caution in renal impairment
Ketorolac	IV	0.2-0.5mg/kg 6-8H	30mg	Recommended duration of therapy to not exceed 5 days *Caution in renal impairment
Mist morphine	Oral	0.2-0.5mg/kg/dose 6 - 8H PRN		Dose applicable to children ≥ 6 mths Do not exceed 5mg per dose when initiating therapy unless APS approves Consult registrar before increasing frequency
Naproxen	Oral	5 – 7mg/kg 8-12H	550mg	Use in children ≥ 2 yo Tab: 275mg Max dose/day: 1100mg *Caution in renal impairment
Tramadol	Oral/IV	1-2mg/kg/dose 6H PRN	100mg	Avoid use in < 12 yo Do not exceed 50 mg/dose for <50kg; max daily dose 400mg without Acute Pain Services (APS) consult. *Causes nausea
CONSTIPATION				
Bisacodyl	Oral	3-12yo: 5-10mg OD >12yo: 10mg OD	10mg	Tablets not recommended for child <3yo. Tab: 5mg (Cannot be crushed)
	Rectal	2-10yo: 5-10mg OD >10yo: 10mg OD	10mg	
Glycerin	Rectal	1-6 mo: 1g (0.5 supp) >6mo to 2yo: 2g (1 supp) >2yo to 6yo: 3g (1.5 supp)		Each suppository = 2g
Fleet® Enema	Rectal	≥ 2 – 12yo: 1 bottle Paeds enema ≥12yo: 1 bottle Adult enema		Not recommended for use < 2 yo Paeds enema – 2.3oz (66.6mL) Adult enema – 4.5oz (133mL) *Caution in CKD: PO ₄ content
Lactulose	Oral	Laxative: 0.5mL/kg 12-24H Hepatic Coma: 1mL/kg 1H, then 6H		

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CONSTIPATION (Continued)				
Macrogol/ Polyethylene Glycol 4000 (Forlax®)	Oral	Maintenance: 0.4 – 0.8 g/kg/day Disimpaction: 1 – 1.5 g/kg/day (Max dose 8 sachets for disimpaction) *Dissolve each sachet in ≥120mL water	2 sachet	Dose applicable to children ≥ 6 mths 1 sachet = 10g Round off to nearest 5g (1/2 sachet) For disimpaction protocol, refer to constipation guidelines on intranet
Sennosides	Oral	2-6yo: 0.5-1 tablet ON 6-12yo: 1-2 tablets ON	15mg	Tab: 7.5mg
ANTI-DIARRHOEALS				
Lacteol fort	Oral	≤2yo: ½ sachet BD >2yo: 1 sachet BD		<i>Lactobacillus acidophilus</i> strain Avoid in immunocompromised patients
LactoGG	Oral	<1yo: 0.5 capsule OD ≥1yo: 1 capsule OD		<i>Lactobacillus rhamnosus</i> GG strain Avoid in immunocompromised patients
Racecadotril (Hidrasec®)	Oral	<9kg: 10mg TDS 9-13kg: 20mg TDS 13-27kg: 30mg TDS >27kg: 60mg TDS		Not recommended for use < 3 months; avoid using >5 days Sachet size: 10mg, 30mg
Smecta	Oral	>2yo: 1 sachet BD-TDS		Not recommended in ≤2yo due to lead content
ABDOMINAL PAIN: GASTRITIS				
Famotidine	Oral	<3mo: 0.5-1mg/kg OD ≥3mo: 0.5-1mg/kg BD	40mg	Tab: 20mg Can increase to 1mg/kg/dose If not effective after 2 weeks
	IV	<3mo: 0.25mg/kg OD ≥3mo: 0.25-0.5mg/kg BD	20mg	
1. Gaviscon® OR 2. MMT	Oral	6-12yo: 5-10ml TDS PRN >12yo: 10-20ml TDS PRN		Give after meals/bedtime SE: diarrhoea 10ml contains 6.2mmol of sodium – avoid in sodium restricted diet
Esomeprazole	Oral	<20kg: 10mg OD ≥20kg: 20mg OD	40mg	Tab: 20mg, 40mg
	IV	Infants: 0.5mg/kg OD Children and Adolescents: <55kg: 10mg OD ≥55kg: 20mg OD	40mg	*Consult GI if requiring higher dosing at BD
Omeprazole	Oral	0.5-1mg/kg/dose OD-BD	40mg	Susp 2mg/ml (expensive compared to capsule), Cap: 20mg
	IV	1mg/kg/dose 8-12H	40mg	
ABDOMINAL PAIN: COLIC				
Ridwind®	Oral	Infants and children <2yo or <11kg: 0.2ml QDS 2 – 12yo (>11kg): 0.4ml QDS		Each 1ml = 100mg Simethicone Apply to bottle or directly to mouth Max 2.4ml/day
Trimebutine (Debridat®)	Oral	<1yo: 25mg BD 1 – 5yo: 25mg TDS 5 - <12yo: 50mg TDS ≥12yo: 100-200mg TDS	200mg	Tab: 100mg (Rationalize to nearest 25mg)
ANTI-EMETIC				
Cinnarizine	Oral	0.3-0.6mg/kg TDS	25mg	Not to be used in <5 yo Tablet = 25mg
Domperidone	Oral	0.2-0.5mg/kg 8H	10mg	Susp: 1mg/mL, Tab: 10mg *Watch for QT prolongation
Granisetron	Oral/IV	0.02-0.04mg/kg Q8H	3mg	For use in Paediatric Oncology Tablet = 1mg
Ondansetron	Oral/IV/ SL	0.1-0.2mg/kg 8H	8mg	For use in children > 6months Higher doses of 6mg/m ² /dose, or 6H dosing frequency for chemo
Promethazine	Oral/IM	0.25-0.5mg/kg 6-8H	10-25mg	Not to be used in < 6yo

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MUCOLYTIC: *Avoid in <2 years old.				
<i>For 2-6 years, trial as second line therapy AFTER supportive care. Discontinue if no improvement by 3 days</i>				
Acetylcysteine (Fluimucil®)	Oral	2 – 6yo: 100mg BD >6yo: 200mg BD <u>OR</u> 600mg OD	600mg	1 sachet = 100mg N-Acetylcysteine 1 effervescent tab = 600mg
Bromhexine	Oral	2 - 6yo: 2mg (2.5ml) TDS 6 - 12yo: 4mg (5ml) TDS >12yo: 8mg (10ml) TDS	8-16 mg	Solution: 4mg/5ml, Tab: 8mg
COUGH SUPPRESSANTS				
Dextromethorphan	Oral	2-5yo: 2.5mL TDS 6-12yo: 5mL TDS >12yo: 5-10mL TDS	10mL	Solution: 15mg/5mL
Promethazine	Oral	6 -12yo: 5-10mg BD >12yo: 10-15mg BD	25mg	Solution: 5mg/5ml Not to be used in <6 yo Avoid concomitant antihistamines
NASAL DECONGESTANTS				
Oxymetazoline (Iliadin®)	Nasal	*<1mo: use NS drops instead 1mo-1yo: 0.01% 1 drop TDS 1-6yo: 0.025% 1 drop TDS >6yo: 0.05% 1 drop TDS		Duration ≤ 5 days to avoid rebound congestion 0.05% available as both nose drop and nasal spray
ANTI-HISTAMINES * Avoid use of ≥ 1 antihistamine				
Cetirizine (Zyrtec®)	Oral	6mo-2yo: 2.5mg OD 2- <6yo: 2.5-5mg OD ≥6yo: 5-10mg OD	10 mg	Solution: 1mg/mL Tab: 10mg
Chlorpheniramine	Oral	2-5yo: 1mg TDS-QDS 6-11yo: 2mg TDS-QDS 12-17yo: 4mg TDS-QDS	4 mg	Sedating antihistamine Solution: 4mg/5mL Tab: 4mg
Fexofenadine (Telfast®)	Oral	6mo - <2yo & 10.5kg: 15mg BD 2-11yo: 30mg BD ≥12yo: 60mg BD <u>OR</u> 180mg OD	180mg	Suspension: 30mg/5mL Tab: 180mg
Hydroxyzine	Oral	>6mo: 0.5mg/kg/dose TDS-QDS <6yo: max 12.5mg/dose >6yo: max 25mg/dose	25mg	Sedating antihistamine Solution: 10mg/5mL Tab: 10mg, 25mg
Diphenhydramine	IV	0.5-1 mg/kg 6-8H	50mg	Sedating antihistamine
ASTHMA				
Salbutamol	MDI	2-6 puffs 4-6H	8 puffs	Via spacer and mask
	Neb	<5yo: 0.5ml 0.5% salbutamol: 2.5ml 0.9% NaCl ≥5yo: 1ml 0.5% salbutamol: 2ml 0.9% NaCl	1ml	Minimum salbutamol volume 0.5mL Up to every 15 mins x 3 doses and to be reviewed by doctor
Ipratropium Bromide	MDI	<6yo: 2-4 puffs 6-8H ≥6yo: 4-8 puffs 6-8H		Via spacer and mask 1 puff = 20mcg
	Neb	1ml Ipratropium bromide with 0.9% NaCl 6-8 hourly	1ml	Nebulizing solution 250mcg/mL (0.025%)
Combination Neb: Salbutamol + Ipratropium + NaCl	Neb	<5yo: 0.5ml 0.5% salbutamol: 1mL Ipratropium: 2.5ml 0.9% NaCl ≥5yo: 1ml 0.5% salbutamol: 1ml Ipratropium: 2ml 0.9% NaCl		Allow max of 1 cycle (Q15mins x 3)
Prednisolone	Oral	1-2mg/kg OD for 3-5 days	40mg	No need taper dose if <14 days
Hydrocortisone	IV	Asthma: 4mg/kg 6H	100mg	
Magnesium SO4 50% (2mmol/mL)	IV	0.2mmol/kg in 10ml 0.9% NaCl over 20 minutes 6H	4ml = 8mmol	Only for status asthmaticus (NOT Mg replacement). Monitor for hypotension
NEBULISED THERAPIES				
Salbutamol + 3% Hypertonic Saline	Neb	0.5ml 0.5% salbutamol: 0.5ml 20% saline: 3ml 0.9% NaCl 4-8H		*NOT for routine use in Bronchiolitis. Give and assess: continue 4-8H ONLY if good response and well tolerated

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NEBULISED THERAPIES (Continued)				
Hypertonic Saline	Neb	3% (Mucoclear) – 4ml (no dilution required) as required 7% - 1ml NaCl 20%: 2ml NaCl 0.9% as required		For 3%: 0.5mL NaCl 20% + 3.5mL NaCl 0.9%
CROUP				
Dexamethasone	Oral	Mild: 0.15mg/kg/dose Moderate to Severe: 0.6mg/kg/dose	10mg	High dose 0.6mg/kg used as single dose (can be given IV/IM/PO) Up to 3 doses MAX, 12hrs apart
Budesonide	Neb	2mg as a single dose, or 1mg for 2 doses separated by 30min interval	0.5mg	May repeat every 12H until clinical improvement Use 1mg/2ml respule; NO need dilute
Adrenaline	Neb	0.5ml/kg of 1:1000 (1mg/ml, 0.1%) Repeat after 30min if necessary	5ml = 5mg	Severe croup; watch for rebound. Dilute with NaCl 0.9% or give neat Effects: 2-3hrs
ANAPHYLAXIS				
Adrenaline	SC/IM	0.01ml/kg of 1:1000 (1mg/mL, 0.1%)	0.5ml	*NOT to be given intravenously
SEIZURES				
Diazepam	IV	0.1-0.4mg/kg	10-20mg	Pls state PRN in instructions when ordering. Repeat after 10min if seizing. Up to max 2 doses of BZD (rectal/IV)
	Rectal	5 - 10kg: 2.5mg >10kg: 5mg >20kg: 10mg	20mg	One rectal tube: 5mg/2.5mL For use in ≥ 5kg, use with caution in <6 months old
Lorazepam	IV	0.05-0.1mg/kg Repeat after 10min if seizing	4 mg	Up to max 2 doses of benzodiazepines (rectal/IV)
Midazolam	IM	0.2mg/kg	10mg	ONLY if IV access is difficult, in status epilepticus
Levetiracetam	Oral/IV	Loading dose: 20 - 40mg/kg Maintenance dose: 10mg /kg 12H (starts 12hrs post-loading)	3g	
PhenOBARBITONE	Oral/IV	Loading dose: 20mg/kg Sub bolus if still seizing: 5mg/kg Maintenance dose: 2.5-3mg/kg 12H (starts 12hrs post-loading)		Observe for resp depression Requires cardiac monitoring Dilute dose with 10x the volume Max infusion rate 1mg/kg/min
PhenYTOIN	Oral/IV	Loading dose: 15 - 20mg/kg Maintenance dose: 2.5-3mg/kg 12H (starts 12hrs post-loading) *aim trough levels 10 to 20		Requires cardiac monitoring Risk of extravasation – need large plug and vein with 0.22micron filter Max con for infusion is 5mg/ml Max infusion rate 1mg/kg/min
ANTI-HYPERTENSIVES				
Amlodipine	Oral	0.1-0.6mg/kg 24H	10mg	
Nifedipine	Oral	0.04-0.25mg/kg 4-6H	10mg	Immediate release tablet Acute Rx, not for long term BP control
Fruzemide	Oral/IV	0.5-1mg/kg 6-24H	80mg	Monitor potassium level
CEREBRAL EDEMA				
Dexamethasone	IV	0.5-2mg/kg STAT (max 10mg) THEN 0.25mg/kg 6H	4mg	Space-occupying lesion *NOT for traumatic brain injury
3% NaCl	IV	5ml/kg		Infuse over 10-20 minutes
20% Mannitol	IV	5ml/kg	500ml	Infuse over 20-30 minutes
CORTICOSTEROIDS				
Hydrocortisone	IV	25mg/m ² 6H (100mg/m ² /day)	100mg	Physiologic stress dose/ Peri-invasive procedures for adrenal/ ACTH insufficiency
	IV	1mg/kg 6H	100mg	Catecholamine-resistant shock
CORTICOSTEROIDS Conversion Ratios (mg) – per DAY equivalent				
Hydrocortisone		Prednisolone	Dexamethasone	Methylprednisolone
4		1	0.15	0.8

**DEPARTMENTAL DRUG REFERENCE
COMMON PAEDIATRIC DRUGS ON CALL (8TH EDITION)**

DRUG	ROUTE	PAEDIATRIC DOSE	ADULT DOSE	REMARKS
HYPOGLYCAEMIA				
Dextrose 10%	IV	2mL/kg STAT	If child is alert and able to feed, consider oral rescue with simple, followed by complex carbohydrate	
COAGULOPATHY				
Phytomenadione (Vitamin K)	Oral/IV	0.3mg/kg	10mg	Oral Vit K given using parenteral vial Slow bolus: Max 1mg/min
INTUBATION				
Atropine	IV	0.02mg/kg	0.6mg	
Etomidate	IV	0.2-0.4mg/kg	20mg	Safe with hemodynamic instability, neuroprotective, transient adrenal cortico-suppression. Avoid in septic shock.
Fentanyl	IV	1-2mcg/kg	50mcg- 100mcg	Caution in patient with hemodynamic instability
Ketamine	IV	1-2mg/kg	50-100mg	Safe with hemodynamic instability if patient is not catecholamine depleted. Use with caution in patients with bronchospasm and septic shock. May cause laryngospasm and increase secretions Avoid in open globe injury.
Midazolam	IV	0.2-0.3mg/kg	5mg	Caution in patients in pre-shock, hemodynamic instability or have depressed cardiac function.
Rocuronium	IV	0.6-1.2mg/kg	50mg (1 vial)	Onset: 30s-1min. Duration: 26-40mins
Suxamethonium	IV	2mg/kg		Contraindications: rhabdomyolysis, chronic skeletal muscle disease or denervating neuromuscular disease, burns, hyperkalaemia, personal/family history of malignant hyperthermia and renal failure Onset: <1min, watch for fasciculations. Duration: 4-10mins
SEDATION				
Chloral Hydrate	Oral	50-65mg/kg	1g	May repeat dose after 30 mins if necessary Max 100mg/kg or total 2000mg/procedure Use with caution in renal dysfunction
Fentanyl	IV	1-2mcg/kg	50-100mcg	To start with 1mcg/kg
Ketamine	IM	3-4mg/kg		
	IV	1-2mg/kg	200 mg	
	Intranasal	3-5mg/kg/dose (half the dose/ per nostril)		Use with MAD (metered atomizer device)
Midazolam	IV	0.05-0.5mg/kg	5mg	Start low; Maximum cumulative dose of 0.5mg/kg; avoid exceeding 5mg
REVERSAL AGENTS FOR SEDATION				
Naloxone	IV	Post-sedation: 0.002mg/kg/dose (0.4mg diluted to 20ml, give 0.1ml/kg/dose)	Max total dose 2mg	*Antidote for morphine/fentanyl *400mcg/ml *Repeat every 2min x4 doses
		Opiate overdose: 100mcg/kg/dose	Max total dose 2mg	*Antidote for morphine/fentanyl *400mcg/ml *Repeat every 2-3min until desired level of reversal
Flumazenil	IV	5mcg/kg every 1min until awake	Max total dose 2mg	*Antidote for benzodiazepines (Midazolam/ Diazepam) *100mcg/ml *BEWARE the use of Flumazenil in patients on benzodiazepines for seizure control

DRUG PRODUCT	ROUTE	PAEDIATRIC DOSE	ADULT DOSE	REMARKS
		INDICATIONS	PAEDIATRIC DOSE	REMARKS
BLOOD PRODUCTS				
Packed Cells (leucocyte reduced)		Anemia	10 -15ml/kg over 3-4H	To review at end of transfusion for IV frusemide (1mg/kg) Order irradiated PCT for pts on <u>clofarabine, stem cell transplant, GVHD, cellular immunodeficiency</u> 4ml/kg of PCT increases Hb by 1g/dL
Platelets		Thrombocytopenia	10ml/kg (0.1unit/kg) free flow	1 unit of platelets will increase plt by 10,000/uL 1 Cell Separator Platelets (CSP) = 4 units of plts
Fresh Frozen Plasma		Coagulopathy/DIVC, Coag Factor def, Warfarin toxicity	10-20ml/kg over 2-4H	To be used within 4hrs of thawing
Cryoprecipitate		Factor VIII def, Hypo/afibrinogenemia (bleeding/ prior to invasive procedures)	5ml/kg or 1 unit/4kg	Contains factor VIII, fibrinogen, vWF, factor XIII Bolus; to be used within 4hrs of thawing
PRE-MEDICATIONS for previous Transfusion Reactions with any Blood Products				
Diphenhydramine	IV	0.5 – 1 mg/kg	Max 50mg	To be given immediately prior to blood product transfusion
Hydrocortisone	IV	1 – 2 mg/kg	Max 100mg	
ORDERING MAINTENANCE INTRAVENOUS FLUIDS FOR A NORMAL CHILD				
A. Type of Maintenance Intravenous Fluid for a normal child				
a. Children <40kg: 5% DSR (Premixed with Potassium Chloride 20mmol/1000ml)				
b. Children ≥40 kg: 5% DSA (0.9% saline with 5% Dextrose + Potassium Chloride)				
B. Calculating intravenous fluid requirement using weight (Holliday-Segar method)				
Weight (kg)	Fluid (ml/day)		Example	
3 – 10	100ml/kg		Volume req for 8 kg child = 8 x 100ml/kg = 800ml/day	
10 – 20	1000ml + 50ml/kg for each kg > 10kg		Volume req for 18 kg child = 1000ml + (50ml/kg x 8) = 1400ml/day	
>20	1500ml + 20ml/kg for each kg > 20kg		Volume req for 28 kg child = 1500ml + (20ml/kg x 8) = 1660ml/day	
Normal max	2500mL			
Fluid requirement of an overweight patient is based on age and ideal body weight (IBW) (kg):				
- Weight for 50% percentile for patient's age OR				
- Formula for IBW (in adol & adults) → Male: 50 + 2.3*(Ht in cm/2.5 - 60) Female: 45.5 + 2.3*(Ht in cm/2.5 - 60)				
*Refer to prescribing guide on intranet for instructions to order drips on Epic				
**PLEASE CONSULT THE DUTY PAEDIATRIC SENIOR DOCTOR WHEN ORDERING FLUIDS FOR NEONATES OR PATIENTS WITH LIVER, RENAL, CARDIAC PROBLEMS OR WITH ELECTROLYTES ABNORMALITIES				
ELECTROLYTES				
SODIUM (HYPONATREMIA – Na < 130mmol/L)				
Sodium deficit (mmol) = 0.6 x body weight x (135 - serum Na)				
Expected change in serum Na per litre of infusate (mmol/L) = (Infusate Na con – serum Na con) / (0.6 x body weight + 1)				
*Should not correct > 10 mmol/L per day unless in acute hyponatremia; please check Na level midway correction				
Oral Sodium			Intravenous Sodium	
o Sodium chloride/dextrose tablet: sodium chloride 500mg and dextrose monohydrate 200mg per tablet = 8.5mmol Na			o 20% sodium chloride: 3.4mmol Na/ml	
o 20% Sodium chloride: 3.4mmol Na/ml			o 3% sodium chloride: 513mmol Na/1000ml	
			o NS/DSA (0.9% NS): 154mmol Na/1000ml	
			o Plasma-Lyte: 140mmol Na/1000ml	
			o DSR (0.45% NS): 77mmol Na/1000ml	
			o DSM (0.23% NS): 38.5mmol Na/1000ml	
WATER (HYPERNATREMIA – Na > 150mmol/L)				
Water deficit over 48 hours (L) = 0.6 x body weight x (1 – serum Na/140)				
Use DSM or DSR				
*Refer to Paeds Renal protocols for details				

ELECTROLYTES (Continued)	
POTASSIUM (HYPOKALEMIA)	
Deficit (mmol) = 0.6 x body weight x (4 - serum K)	
Oral Potassium	Intravenous Potassium (HIGH RISK DRUG)
- Span K: 8.1 mmol K/tab (<i>Inconsistent absorption</i>) - Mist KCl: 1.3 mmol K/ml (<i>Preferable</i>) - Potassium citrate mist: elemental 2.8mmol K/ml, 1.2mmol citrate/mL - Potassium citrate tab: 10mmol K (+ 3.3 mmol citrate) Note: 1 mmol citrate = 3 mmol HCO₃	7.45% Potassium chloride: 1ml = 1mmol K NEVER to be given undiluted - Peripheral vein for IV fluid drip: max 40mmol/L (Ward) and max 80mmol/L (only in PICU) - Peripheral line for acute replacement: max 10mmol per 100mL - Central line for acute replacement: max 20mmol per 100ml - Max rate of K infusion: 0.4mmol/kg/hr (40mmol/hr)
POTASSIUM (HYPERKALEMIA K >5.5mmol/L)	
- ECG monitoring - IV 10% Calcium Gluconate: 0.5 mL/kg (Max 10 mL), given undiluted and bolus over 5 – 10mins - IV 8.4% Sodium Bicarbonate: 1 – 2 mmol/kg (Max 50 mmol/dose) - Resonium A: PO/PR 0.5 – 1 g/kg 6H (Max 60 g/day) [PO max 15g/dose, PR enema max 30g/dose] - Nebulised Salbutamol: Infants and <25kg: 0.5mL Salbutamol + 2.5mL NaCl 0.9% 25 – 50 kg: 1mL Salbutamol + 2mL NaCl 0.9% >50 kg: 2mL Salbutamol + 1mL NaCl 0.9% (Up to 4mL for larger children)	
IV insulin (Actrapid) by slow infusion: 1U insulin with every 5g glucose (1mL/kg of 50% Dextrose (Max 50mL) + 0.1unit/kg (max 10 units) insulin in the same syringe) → To run over 15 mins. If patient hypoglycaemia concurrently, IV fluid drip of 5% DSR must be initiated before IV insulin/dextrose treatment	
BICARBONATE (METABOLIC ACIDOSIS)	
Bicarbonate deficit = 0.3 x (Base deficit) x body weight OR 0.3 x (Ideal – serum HCO₃) x body weight Give half as bolus dose and then infusion over 2 – 4 hours	
Oral Bicarbonate	Intravenous Bicarbonate
- Shohl's solution: Na citrate dihydrate 500 mg, citric acid monohydrate 334 mg per 5ml; 1mmol Na + HCO₃ per ml - Sodium bicarbonate 500mg capsule: 6mmol Na + HCO₃	8.4% sodium bicarbonate: 1mmol of HCO₃/ml ALWAYS USE 4.2% in peripheral vein and in < 1yr old - Intact resp system needed to eliminate CO₂, **iCa will fall
CALCIUM (HYPOCALCEMIA)	
Oral Calcium	Intravenous Calcium
- Calcium liquid (300mg elemental calcium in 15ml): 0.5mmol Ca/mL - Calcium acetate 667mg (25% elemental Ca) tablet: 4.2mmol Ca/tablet; <i>unsuitable for Ca replacement therapy</i> - Calcium carbonate 625mg (40% elemental Ca): 6.3mmol Ca - Calcium carbonate 1250mg (40% elemental Ca): 12.5mmol Ca - Calcium carbonate 450mg/Vit D 200 units: 4.5mmol Ca	- Ca gluconate 10% soln (0.23mmol Ca/ml) :0.5ml/kg (max 20ml in 100ml NS) slow IV infusion over 30-60min/ can be used undiluted and bolus over 3min - Ca chloride 10% soln (0.68mmol Ca/ml): 0.2ml/kg (max 10ml in 50 – 100ml NS) slow IV infusion over 30-60min via central line Note: Ca chloride has 3x elemental Ca compared to Ca gluconate General guide for bigger children (>50kg) - Usual adult 1 cycle = 2.3mmol, give 1 – 2 cycles
PHOSPHATE (HYPOPHOSPHATEMIA)	
Oral Phosphate	Intravenous Phosphate
- Sodium phosphate solution: - Na 1.15mmol and PO₄ 0.7mmol per mL - Phosphate Sandoz effervescent tablet: PO₄ 16.1mmol, Na 20.4mmol, K 3.1mmol and HCO₃ 7.3mmol per tablet *Watch for diarrhea *Caution in using KPO₄ in patients with normal - high K	IV correction only if severe (< 0.32mmol/L) or symptomatic - Potassium dihydrogen phosphate (1mmol/mL PO₄ + K): 0.1-1.5mmol/kg/day (avoid exceeding 70mmol/day) slow IV infusion. Max infusion rate is 0.2mmol/kg/hour - Sodium glycerophosphate (1mmol/ml of PO₄ + 2mmol/mL Na): 0.1-1.5mmol/kg/day (max 70mmol/day) slow IV infusion. Max infusion rate is 0.2mmol/kg/hour - Peripheral: max conc 0.05mmol PO₄/ml (5mmol/100mL) - Central: max conc 0.1mmol PO₄/ml (10mmol/100mL)
MAGNESIUM (HYPOMAGNESEMIA)	
Oral Magnesium	Intravenous Magnesium
- LAC Magnesium 250mg: 10.4mmol Mg/tab - Magnesium sulphate 35% solution: 1.77grams Mg sulphate (7.3 mmol Mg) per 5ml *Watch for diarrhoea	- Magnesium sulphate 50% (2mmol/ml): 0.4mmol/kg slow IV infusion over 30-60 mins (Max conc: 0.8mmol/mL) - For severe hypomagnesemia or renal impairment, please refer to departmental guidelines
General guide for bigger children (>50kg) for Mg replacement: 0.6 – 0.7 mmol/L: IV 10mmol or PO Mg tab 1 – 2 tabs/day 0.5 – 0.59 mmol/L: IV 20mmol or PO Mg tab 1 – 2 tabs/day; <0.5 mmol/L: IV 20 – 30 mmol	

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Disclaimer: Every effort has been made to ensure the accuracy of the information in this guide. The authors will not be responsible for any errors arising or related to its use. Next review in June 2028.

DRUG	ROUTE	PAEDIATRIC DOSE	REMARKS
ANALGESIAS			
Ketoprofen 2.5% Gel	Topical	> 6yo: 1 application BD-TDS PRN	Avoid in patients with NSAIDs allergy
Ketoprofen 30mg Patch	Topical	> 12yo: 1 patch OD-BD PRN	
Diclofenac 1% Gel	Topical	> 14yo: 1 application BD-TDS PRN	
EAR CONDITIONS: OTITIS MEDIA			
Ofloxacin 0.3% Ear Drop (Tarivid Otic)	Topical	1-12yo: 2 drops BD ≥ 12yo: 5 drops BD	Duration: 10-14 days In addition to systemic antibiotics
EAR CONDITIONS: OTITIS EXTERNA			
Ofloxacin 0.3% Ear Drop (Tarivid Otic)	Topical	6mo-12yo: 2 drops BD ≥ 12yo: 5 drops BD	Duration: 7 days
Ciprobay HC Ear Drop (Ciprofloxacin 2mg, Hydrocortisone 10mg/mL)	Topical	≥ 1yo: 3 drops BD [use is contraindicated in suspected / perforated tympanic membrane]	Consider using 1 drop instead in smaller children with smaller ear canal volume
EAR CONDITIONS: OTOMYCOSIS			
Clotrimazole 1% Ear Drop (Candid)	Topical	≥ 12yo: 2 drops TDS	Duration: At least 2 weeks after disappearance of infection
EYE CONDITIONS: INFECTIVE CONJUNCTIVITIS			
Chloramphenicol 0.5% Eye Drop	Topical	1 drop TDS-QDS and continue for 48 hours after healing	Up to 1 drop 2-hourly in severe infections then reduce frequency when infection is controlled. Suitable for infants
Tobradex Eye Drop (Tobramycin 0.3%, dexamethasone 0.1%)	Topical	≥ 2yo: 1-2 drops 4-6H	May be increased to 2H during initial 24-48 hours in severe infection, until improvement
Tobramycin 0.3% Eye Drop (Tobrex)	Topical	1 drop BD-QDS Severe infections 1 drop 4H (Caution about dose-related conjunctival toxicity)	Max 2 drops every hour in severe infection until improvement. *Can be used in neonates, but practice is based on labelled indications
EYE CONDITIONS: ALLERGIC CONJUNCTIVITIS			
Olopatadine HCl 0.1% Eye Drop (Patanol)	Topical	≥ 3yo: 1 drop BD-QDS	
REGULAR DIAPER RASH/ DERMATITIS			
Barrier cream (Conveen Critic Barrier Cream / 3M Cavilon Durable Barrier Cream/Desitin Cream)	Topical	Apply thickly to affected areas	Protective/Barrier cream for excoriated/irritated skin
Drapolene cream (cetrimide/benzalkonium Cl)	Topical	After every diaper change	Antiseptic
Zinc Oxide 40% (Desitin <u>ointment</u>)	Topical	1 application PRN	
FUNGAL DIAPER RASH			
Miconazole nitrate 2% <u>Cream</u>	Topical	1 application BD or 1 application daily when with powder	Duration: 2-6 weeks. Continue for 1 week after symptoms disappear
Miconazole nitrate 2% <u>Powder</u>	Topical	1 application BD or 1 application daily when with cream	Duration: 2-6 weeks. Continue for 1 week after symptoms disappear
Miconazole nitrate 2%, Hydrocortisone 1% Cream	Topical	1 application once to twice daily	Steroid and antifungal combination. May use for 3-4 days until redness improves then switch to anti-fungal monotherapy
ASTRINGENT WASH			
Calamine Lotion	Topical	Apply to affected areas OD-QDS	Anti-pruritic May reapply after bath
Copper-Zinc-Sulphate Solution	Topical/ Wash	1 application BD (As last rinse) Dilute 1:10 parts with water	For eczema (anti-inflammatory) May be used to compress or wash affected areas. Dab dry after use, not necessary to wash off

DRUG	ROUTE	PAEDIATRIC DOSE	REMARKS
ECZEMA / ATOPIC DERMATITIS			
Please refer to KTP-UCMI guidelines for detailed management of eczema with the following agents			
Hydrocortisone 1% Cream	Topical	Apply thinly on affected areas daily to BD	For mild eczema 0.5% strength available for babies/ young infants
Potency of steroid creams			
Mild (Class VI, VII)	Hydrocortisone 0.5% cream, Hydrocortisone 1% cream, Betamethasone valerate 0.05% cream, Desonide 0.05% lotion, Desonide cream 0.05% (Desowen), 0.025% betamethasone valerate cream.		
Moderate (Class IV, V)	Mometasone furoate 0.1% cream/lotion , Hydrocortisone aceponate 0.127% cream, Betamethasone valerate 0.1% cream (≥1yo only) , Fluticasone propionate 0.05% cream		
Potent (Class II, III)	Mometasone furoate 0.1% ointment , Betamethasone dipropionate 0.05% cream/ointment		
Very potent (Class I)	Clobetasol propionate 0.05% cream/ointment (≥1yo only)		
Steroid Sparing Agents			
Pimecrolimus 0.1% Ointment (Elidel®)	Topical	≥3mo: Thinly BD until clearance of lesion	Add-on agent when eczema control is not achieved despite liberal moisturising and use of topical moderate steroids a few times a week. Re-assess factors regarding control → refer to Allergy Team
Tacrolimus 0.03%, 0.1% Ointment (Protopic®)	Topical	2-16yo (0.03%): Thinly BD for 3 wks then OD ≥16yo (0.1%): Thinly BD until clearance of lesion. Attempt to reduce frequency or use 0.03%	
ORAL ULCER (NON-INFECTIVE)			
Detinox Teething Gel (Cetylpyridinium Cl 0.1%, lignocaine HCl 0.33%)	Topical	1 application PRN	Maximum 6x/day (Every 3H)
Oracort E Oral Paste (Triamcinolone Acetonide 0.1%, Lignocaine HCl 3%)	Topical	≥7yo: 1 application BD – QDS	Maximum 4x/day
Soragel/Bonjela (Cetalkonium Cl 0.01%, choline salicylate 8.7%)	Topical	≥ 4mo: 1 application TDS – QDS	Maximum 6x/day (Every 3H)
ORAL ULCER (INFECTIVE)			
Acyclovir 5% cream (Zovirax cream)	Topical	Apply 5 times a day	For herpes labialis Duration: 5-10 days
ORAL CARE / MUCOSITIS			
Chlorhexidine Gluconate 0.2% Mouthwash	Oral	5-10mL BD-TDS PRN	For secondary infection of mucosal ulceration and as adjunct to control gingivitis . Prevention of oral candidiasis in immunocomp.
Glutamine Powder	Oral	0.1g/kg/dose BD-TDS; dissolve in cold/room temperature water; swish and swallow	Prophylaxis/Treatment of chemo/radiation-oral mucositis (Not available inpatient)
Nystatin Suspension (100 000 units/mL)	Oral	<1mo: 1ml QDS 1-12mo: 2mL QDS; cotton/finger to apply ≥12mo: 5mL TDS-QDS	Oral thrush/ candidiasis Duration: 7 days & continue treatment for at least 2 days after symptoms disappear May swish and swallow
OralSeven® Mouthwash	Oral	5-10mL TDS-QDS PRN	Moisturising mouthwash with no alcohol
SORE THROAT			
Dequalinium 0.25mg Lozenges	Oral	≥ 6yo: 1 lozenge TDS PRN	May be a choking hazard Avoid in children with lack of ability to suck on lozenges Avoid Difflam Lozenges in patients with NSAIDs allergy
Difflam Lozenges (Benzylamine HCl 3mg)	Oral	≥ 6yo: 1 lozenge TDS-QDS PRN	
Difflam Throat Spray (Benzylamine HCl 3mg/mL)	Oral	6mo-6yo: 1 spray per 8kg QDS PRN (max 2 sprays/dose) ≥ 6yo: 2 sprays QDS PRN	May be increased to 3H PRN Avoid in patients with NSAIDs allergy

DRUG	ROUTE	PAEDIATRIC DOSE	REMARKS
MRSA TREATMENT / DECOLONIZATION			
Mupirocin 2% Nasal Ointment (Bactroban)	Nasal/ Topical	Apply to the anterior nares of each nostril TDS x 5 days (not longer than 7 days to prevent resistance)	NUH SOP for MRSA decolonization <u>BOTH</u> steps to be done concurrently
Octenisan™ Wash Lotion	Topical	Apply as body wash during showering or wiped over body during sponge bath x 7 days	
SUPERFICIAL ABRASIONS AND WOUNDS			
Baneocin Powder (Bacitracin Zn 250 IU, neomycin sulfate 5,000 IU/g)	Topical	Apply sparingly BD to QDS	For infected abrasions Duration: 7 days
Chlortetracycline 1% eye Ointment	Topical	1 application PRN	For facial region
Fusidic Acid 2% Cream (Fucidin 2% cream)	Topical	1 application BD to TDS	For infected abrasions Duration: 7 – 14 days
Potassium Permanganate Solution (1:1000)	Topical	1 application PRN Dilute 1:8 parts with water	May be used to compress or wash affected areas. Dab dry after use, not necessary to wash off
Tetracycline 3% Ointment	Topical	1 application PRN	Avoid application to facial region for patients <4yo
FIRST DEGREE/ SUPERFICIAL BURNS			
Chlorhexidine Gluconate 1% (Hexodane cream)	Topical	Clean affected area and apply generously	Antiseptic
Drapolene Cream (Benzalkonium chloride 0.01%; cetrimide 0.2%)	Topical	Apply to affected area liberally PRN	Antiseptic. Also, for nappy rash
Silver Sulphadiazine 1% Cream	Topical	>2mo: Apply OD to BD; cover burn area with cream at all times	Broad antimicrobial activity For 2nd-3rd degree burns. Continue applying if there is a possibility of infection
ACNE VULGARIS			
Dalacin T (Clindamycin) 1% Gel	Topical	≥ 12yo: Apply thin film BD	
Differin 0.1% Gel (Adapalene)	Topical	≥ 12yo: Apply ON to affected areas after washing	Avoid mucous membranes; avoid excessive sunlight. Use sunblock.
Epiduo 0.1%/2.5% Gel (Adapalene/benzoyl peroxide)	Topical	≥ 9yo: Apply ON to affected areas	Avoid mucous membranes; avoid excessive sunlight. Use sunblock. Caution in <9yo
Stieva-A (Tretinoin) 0.05% Cream	Topical	≥ 12yo: Apply ON to affected areas for 3 months	Cleanse and dry skin before applying.
DANDRUFF			
Hi-Tar Liquid (Coal Tar 25%)	Topical	Apply on wet hair to produce an abundant lather and soak for 20min. Use 1-2x/week.	Skin will be more sensitive to sun
Ionil T Shampoo (Coal Tar 4.25%/ Salicylic acid 2%)	Topical	Apply small amount into wet hair. Rinse. Apply again & massage into a lather. Soak for 5min. Use 2-3x/week.	For ≥2yo Keratolytic. May take up to 7-10 days to see effects
Ketoconazole 2% Shampoo	Topical	Apply on wet hair to produce an abundant lather and soak for 5 mins. Use 1-2x/week	
SCABIES *(Refer to separate Patient Counselling Leaflet on Intranet)			
Ivermectin	Oral	≥15kg: 0.2mg/kg for 2 doses, 7 days apart	
Malathion 0.5% Lotion	Topical	≥6mo: Apply over whole body (include scalp, neck, face & ears). Wash off after 24hrs. Repeat application after 7 days	Reapply if hands are washed with soap within 24hrs
Permethrin 5% Lotion	Topical	≥ 2mo: Apply over whole body including face, neck, scalp and ears; wash off after 8–14hrs. Repeat application after 7 days	Reapply if hands washed with soap within 8hrs of application

DRUG	ROUTE	PAEDIATRIC DOSE	REMARKS
HEAD LICE			
Malathion 0.5% Lotion	Topical	≥ 6mo: Apply sufficient amount on dry hair to thoroughly wet the hair and scalp and allow the hair to dry naturally. Wash after ≥8-12hrs. Repeat application after 7-9 days if needed	Use a fine-toothed (nit) comb to remove dead lice and eggs.
ANTHELMINTIC			
Albendazole	Oral	Pin-/Hook-/Whip-/Round worm infection: 1-2yo: 200mg single dose (premeal) >2yo: 400mg single dose (premeal) Max: 400mg single dose	Tab: 400mg May use up to 3-5 consecutive days depending on indication
SUPPLEMENTS			
Appeton Multivitamin Infant drops (30 mL)	Oral	0-12mo: 1mL OD	Contains Vit A, B (1, 2, 6, 12), C, D3, E, Taurine, Lysine
Wellkid Multivitamin Liquid (150 mL)	Oral	1-10yo: 5mL OD >10yo: 10mL OD	Contains Vit A, B (1, 2, 3, 5, 6, 12), D3, C, E, Folic, zinc, copper, iron 5mg/5mL
DEKAS Plus Liquid (60 mL)	Oral	0-12mo: 1mL OD 1-3yo: 2mL OD	Contains Vit A, B (1, 2, 3, 5, 6, 7), C, D3, E, K, CoQ10, Selenium, Zinc
Ferric Hyd Polymaltose Drops (50 mg/mL)	Oral	3-6mg/kg/day elemental iron	50mg elemental iron/mL 1mL = 20 drops
Iron Tablets	Oral	1. Iron Polymaltose 100mg Capsule 100% elemental iron (100mg) 2. Ferrous Gluconate Cap (Sangobion) 12% elemental Fe = 30mg elemental Fe (also contains cyanocobalamin, folic acid, ascorbic acid, manganese, copper sulphate, sorbitol)	3. Iberet Folate SR Tab 20% elemental Fe = 105mg elemental Fe (also contains thiamine, pyridoxine, cyanocobalamin, riboflavin, niacinamide, folic acid, calcium panthothenate, ascorbic acid)
Folic Acid	Oral	1mo - <1 yo: 400-800mcg OD 1 - 5yo: 2.5mg OD ≥5yo: 5mg OD	Strengths available: - 400mcg tablets - 5mg tablets
Vitamin D <i>Refer to NUHkids paediatric renal guidelines for dosing in patients with CKD</i>	Oral	Recommended Dietary Allowance: <12mo: 400IU OD ≥12mo: 400-600IU OD Vitamin D insufficient or deficient: <12mo: 2000 IU OD for 90 days 12mo – 12yo: 3000-6000 IU OD for 90 days >12yo: 6000 IU OD for 90 days Vitamin D stoss dosing: 3-12mo: 50 000 IU (2mL) once 12mo-12yo: 150 000 IU (6mL) once >12yo: 300 000 IU (12mL) once	Formulations available: - Tablet (1000IU per tab) - Solution (25 000IU/vial) - Drops (200IU per drop)
COMMON DRESSINGS FOR SUPERFICIAL SKIN LESIONS			
You may refer to NUH Pharmacy guidelines (under drug list) for details of the individual dressings. Please kindly refer to the appropriate expertise for complicated wounds/skin lesions			
Types of Wound	Recommended Dressing		
Superficial Acute /Traumatic	Foams (Allevyn), Silicon Foam (Mepilex XT / lite), Hydrocolloids (Duoderm), Hydrofibre (Aquacel), Lipido-colloid (Urgotul), Silver (Mepilex Ag)		
Bleeding	Alginates (Kaltostat), (Biatain)		
Exudating and Infected	Alginates (Kaltostat), (Biatain) , Hydrofibre (Aquacel), NaCl dressing (Mesalt)		
Heavily Exudating	Alginates (Kaltostat), (Biatain) , NaCl dressing (Mesalt), Lipido-colloid (UrgoCell), Silver (Aquacel Ag)		
Light to Mod Exudating	Foam (Allevyn), Silicone Foam (Mepilex XT/ Mepilex lite), Hydrocolloids (Duoderm CGF/Extra)		
Burst Blister	Hydrocolloids (Duoderm CGF/Extra)		
Sloughy	Hydrogels (Duoderm gel/Purilon gel), Alginates (Kaltostat), (Biatain)		
Dry with Scab or Necrosis	Hydrogels (Duoderm gel/Purilon gel)		
Secondary Healing	Silicone Foam (Mepilex XT/ Mepilex lite) (Allevyn)		
Superficial Burns	Silicone Foam (Mepilex XT/ Mepilex lite)		

PAEDIATRIC ENTERAL FEEDS							
Medical Condition	Age Range	Possible Feeds	Standard Dilution	Type of feed	Scoop Size (per scoop)	Flavour	Remarks
Failure To Thrive (FTT) *Children aged 0-1 years may benefit from dietitian input to concentrate up milk feeds	Premature	Neosure	0.74kcal/ml	Powder	9.8g	Unflavoured	
	≥1 yr old	Nutren Junior	1kcal/ml	Powder	7.86g	Vanilla	Contains probiotics - not for post-transplant patients
	≥1 yr old	Pediasure	1kcal/ml	Liquid	220ml/bottle	Vanilla	Almost lactose free No pro-biotics For post-transplant patients
	>4 yrs old	Resource fruit beverage	1.5kcal/ml	Ready to Drink	200ml/packet	Apple	Milk free
		Ensure Life StrengthPro	1.2kcal/ml	Ready to Drink	220ml/bottle	Vanilla	Lactose free
		Resource Protein chocolate	1.25kcal/ml	Ready to Drink	200ml/packet	Chocolate	Lactose free
		Fresubin Energy	1.5kcal/ml	Ready to Drink	200ml/bottle	Tropical fruits	Low lactose
		Resource 2.0	2kcal/ml	Ready to Drink	200ml/packet	Vanilla	Lactose free
Cow's Milk Protein Allergy	0-36 mths	Alfare	0.68kcal/ml	Powder	4.5g	Unflavoured	Extensively hydrolysed; 40% fat as MCT May be used for liver patients/ short bowel syndrome/ protein-losing enteropathy
Short Bowel Syndrome or Protein-losing Enteropathy Or Multiple food allergies	0-12 mths	Neocate LCP	0.67kcal/ml	Powder	4.5g	Unflavoured	Elemental/amino acid No MCT oil Multiple food protein allergy
		Comidagen	0.68kcal/ml	Powder	4.5g	Unflavoured	
	>1 yr old	Comidagen Plus	1kcal/ml	Powder	7.3g	Unflavoured	Elemental/amino acid No MCT oil Multiple food protein allergy
		Peptamen Junior	1kcal/ml	Powder	7.9g	Vanilla	
Chylothorax	Birth to 10 years	Lipistart	0.7kcal/ml	Powder	5g	Unflavoured	74% of fat as MCT oil May be used for patients with intestinal lymphangectasia
Renal Failure	Birth to 10 years	Renastart	1kcal/ml	Powder	7g	Unflavoured	Low in electrolytes PO ₄ , K
	>4 yrs old	Nepro LP	1.8kcal/ml	Ready to Drink	220ml/bottle	Vanilla	10g protein/serving Lactose & gluten free Low in electrolytes Na, PO ₄ , K
		Nepro HP	1.8kcal/ml	Ready to Drink	220ml/bottle	Vanilla	18g protein/serving Lactose & gluten free Low in electrolytes Na, PO ₄ , K
Crohn's Disease	≥5 yrs old	Modulen IBD	1kcal/ml	Powder	8.3g	Neutral	Nutritionally complete 25% fat as MCT Lactose free
Seizures/ Epilepsy	≥1 yr old	Ketocal (4:1)	1kcal/ml	Powder		Unflavoured	For patients on Ketogenic Diet (KD)
	≥3 yrs old	Kflo (4:1)	1.5kcal/ml	Liquid	250ml/bottle	Vanilla	For patients on Ketogenic Diet (KD)
Modular Component	Any age	Valens Myotein	-	Powder	6.3g	Unflavoured	Instantised Whey protein Gluten free Contains lactose, trace amounts of carbohydrate, fat, Ca, Na, K, Mg, Phosphorus
		Valens Carborie	-	Powder	2g	Unflavoured	Readily absorbed glucose polymer Gluten and lactose free Contains trace amounts of Ca, Na, K, Cl, phosphorus
		Enersos MCT oil	-	Liquid		-	100% pure MCT oil , contains trace amounts of Ca, iron, Na

NATIONAL CHILDHOOD IMMUNIZATION SCHEDULE

Vaccine	Birth	2 months	4 months	6 months	12 months	15 months	18 months	2-4 years	5-9 years	10-11 years	12-13 years	13-14 years	15-17 years
Bacillus Calmette-Guérin (BCG)	D1												
Hepatitis B (HepB)	D1	D2		D3									
Diphtheria, tetanus and acellular pertussis (paediatric) (DTaP)		D1	D2	D3			B1						
Tetanus, reduced diphtheria and acellular pertussis (Tdap)										B2			
Inactivated poliovirus (IPV)		D1	D2	D3			B1			B2			
<i>Haemophilus influenzae</i> type b (Hib)		D1	D2	D3			B1						
Pneumococcal conjugate (PCV13)			D1	D2	B1								
Pneumococcal polysaccharide (PPSV23)								One or two doses for children and adolescents age 2-17 years with specific medical condition or indication.					
Measles, mumps and rubella (MMR)					D1	D2							
Varicella (VAR)					D1	D2							
Human papillomavirus (HPV2)											D1 (Females)	D2 (Females)	
Influenza (INF)					Annual vaccination or per season for <u>all children</u> age 6 months to <5 years (6-59 months).				Annual vaccination or per season for children and adolescents age 5-17 years with specific medical condition or indication.				

 Recommended ages and doses for all children

 Recommended for persons with specific medical condition or indication

Footnotes for NCIS:

- **D1, D2, D3:** Dose 1, dose 2, dose 3
- **B1, B2:** Booster 1, booster 2
- **10-11, 12-13, 13-14 years:** Pri 5, Sec 1, Sec 2 (Tdap, IPV, HPV (for females) and MMR (as catch-up) vaccines are provided as part of HPB's school-based vaccination programme)
- **HepB:** Doses 2 and 3 are recommended to be given as part of the 6-in-1 vaccine at 2 and 6 months, respectively
- **MMR:** Only dose 2 is recommended to be given as part of MMRV vaccine

Notes on Vaccines:

- BCG** - Ideally given at birth. If to be given beyond 6 mths, do PPD first to check if already exposed.
- Hep B** - The birth dose may be used as the 1st dose in a 3 dose primary series with 4 wk intervals. Neonates born to HBsAg+ mothers: 3 doses at 0, 1 (give 5-in-1 at 2 mths), 6 mths (as 6-in-1).
- DTP** - 2nd Booster for DTP is given as reduced antigen Tdap.
- Rotavirus** – Options: Rotarix, RotaTeq. 1st dose to be given between 6-14 wks. Minimum interval 4 weeks. Rotarix: 2 doses, 2nd dose by 24 wks. RotaTeq: 3 doses, last dose by 32 wks.
- Pneumococcal** – If 2 doses given ≥12mths, or 1st dose given 24-59mths, no further doses needed.
- HPV** – Cervarix (HPV2) / Gardasil (HPV4) / Gardasil 9 (HPV9). Females 9-14YO: HPV2/HPV4: 2-doses at 0, 6 mths. Females 15-17YO: HPV2: 3-doses at 0, 1, 6 mths, HPV4: 3-doses at 0, 2, 6 mths.
- Influenza:** 6MO - <9YO: 2-dose series 4 weeks apart for those receiving for the first time. Otherwise annually or per season as recommended.

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VITAL PARAMETERS

Estimated Weight = $[2 \times (\text{age in years} + 4)] \text{ kg}$
Lower limit Systolic BP = $[70 + (\text{age in years} \times 2)] \text{ mmHg}$
Diastolic BP = $2/3^{\text{rd}}$ systolic BP
BSA (m²) = $\sqrt{(\text{Height cm} \times \text{Wt kg} / 3600)}$
Insensible Fluid Loss (ml) = $400\text{ml/m}^2 \times \text{BSA (m}^2\text{)}$

Age	RR (Breaths/min)	HR (Bpm)
0 - 3M	30-60	80-160
3M - 1Y	30-45	75-160
1Y - 3Y	25-35	75-140
3Y - 6Y	20-30	75-120
6Y - 10Y	16-25	70-110
10Y - 15Y	14-20	60-100
>15Y	12-16	60-90

	Recommended	eGFR Equation
Pre-term infants (0-12 months)	Revised Schwartz (Adjusted)	$22.1 \times \text{L/Cr}$
Full-term infants (0-12 months)	Revised Schwartz (Adjusted)	$30.1 \times \text{L/Cr}$
Child 1-12 years	Revised Schwartz	$36.5 \times \text{L/Cr}$
Adolescents 12-17 years	Revised Schwartz	$36.5 \times \text{L/Cr}$
Adult ≥ 18 years [#]	CKD-EPI	$141 \times \min \left\{ \left\{ \frac{\text{Cr}}{(k \times 88.4)} \right\}, 1 \right\}^{\alpha} \times \max \left\{ \left\{ \frac{\text{Cr}}{(k \times 88.4)} \right\}, 1 \right\}^{-1.209} \times 0.993^{\text{age}} \times 1.018^{\text{(if female)}}$
Patients with abnormal muscle mass or body habitus	Cystatin C levels	
Acute kidney injury	Cystatin C levels	

eGFR: In ml/min/1.73m²

L : Length or height in cm

Cr : Serum creatinine in $\mu\text{mol/L}$

k : Constant (0.7 for females and 0.9 for males)

α : Constant (-0.329 for females and -0.411 for males)

min : Minimum

max : Maximum

[#] : Consider average of Revised Schwartz + CKD-EPI equations for 18-26 years

Laboratories		Radiology		Clinics / CE / Wards	
Haematology	22323 / 25354 / 25320	Portable XR Office hrs:	9820 9950	KTP-NUCMI	25736
		After hrs:	9670 7437	EEG Rm	24123
		DDI	25201	2D Echo Rm	24138
Blood Bank	22305	EMD Radiographer	97220502	Lung Fxn Rm	24450
BTS MO	9186 4133	CT (PSA / Control)	22211 / 25396	Day Therapy	27814
Biochemistry	24346 / 22993	MRI (PSA / Control)	25360 / 25364	BMU	25700
Microbiology	24343 / 22312	KRW MRI	26673 / 24775	CE senior phone	9088 3346
MDC	24384 / 24175	MCU / Fluoroscopy	22091	CE (main / FF)	11837 / 24995
Pathology Admin:	22330 / 22332	Ultrasound	22212	PICU / HD	25459 / 25460
Lab:	24311	IR (PSA / Nursing)	22216 / 24218	W45 / CKC	25450 / 22405
Paeds Lab	6601 3306	BMD	22207	W47 / 47A	25470 / 25478
Siew Lan / Puva	24912	Nuclear Med	24212	W62	25620
Pharmacy		PET-CT	22351	W9B	26660 / 26661 (Front) 26628 / 26629 (Back)
Main Pharm	24506 (Lvl 5)	MSW Office	25167	W8B / BMT	25880 / 25899
KRW Pharm	25751 (Lvl 7)	Paeds Admin / PC's Office		VUC3 front desk	25030
Sok Hoon	8428 1395	Serena (MO/Res)	23392	VUC3 treatment	25898 / 13833
Cheryl Neoh	9155 6242	Pearlene (MO/Res)	22032	OT (Recovery)	24900
Louis Lim	9773 0163	Jenny (PGY1)	22604	OT (Reception)	22911
Pharm on call	9723 6471				

USEFUL NEONATAL REFERENCES

Serum Bilirubin (SB) levels (umol/L) in **Standard Risk** Infants

AGE	TcB LEVELS	SINGLE BLUE PHOTOTHERAPY	DOUBLE BLUE PHOTOTHERAPY	EXCHANGE TRANSFUSION
0-12h	Manage as per High Risk Criteria			
13-18h				
19-24h	135	150	180	250
25-36h	160	180 Not for discharge >160 OR weight loss >90 th or 6% https://www.newbornweight.org/chart/	250	300
37-48h	200	225	275	350
49-72h	225	250	300	350
73-96h	250	275	325	375
97-120h	250	275	350	400
121h-7 days	270	300	350	400
8-14 days	≥300	325	375	425

Serum Bilirubin (SB) levels (umol/L) in **High-Risk*** Infants

AGE	TcB LEVELS	SINGLE BLUE PHOTOTHERAPY	DOUBLE BLUE PHOTOTHERAPY	EXCHANGE TRANSFUSION
0-12h	90	-	100	200
13-18h	100	120	150	250
19-24h	120	130	150	250
25-36h	135	150	200	275
37-48h	160	180	225	300
49-72h	200	225	275	325
73-96h	225	250	300	350
97-120h	225	250	300	375
121h-7 days	250	275	325	375
8-14 days	270	300	325	375

***Criteria for High-Risk Neonatal Jaundice:**

- **G6PD deficiency & other haemolytic anaemia (Refer to G6PD threshold and guidelines)**
- **ABO incompatibility** (defined as maternal blood group O and baby blood group A, B or AB AND DCT positive OR Reticulocyte count > 8% or rapid rise in serum bilirubin > 103 umol/L per day)
- **Rhesus incompatibility**
- **Rapid rise in serum bilirubin > 103 umol/L per day**
- **Family history of severe jaundice** (e.g. siblings need exchange transfusion)
- **History of prematurity**

USEFUL FORMULAE IN INTENSIVE CARE

Calculation of Glucose Infusion Rate (GIR)	Formulae pertaining to ETT*
$\frac{\text{Dextrose Concentration (\%)} \times \text{Infusion Rate (ml/hr)}}{\text{Weight (kg)} \times 6}$	Uncuffed ETT diameter (mm): (Age / 4) + 4 Cuffed ETT diameter (mm): (Age / 4) + 3.5 Depth of ETT (Orotracheal intubation): Size of ETT x 3 (cm)

*Refer to Broselow Tape for estimated sizes if age unknown

QUICK REFERENCE GUIDE FOR PAEDIATRIC EMERGENCIES / CALL CODE BLUE *27

AGE			NEWBORN	1 MONTH	3 MONTHS	6 MONTHS
WEIGHT			3 KG	4 KG	5 KG	8 KG
	Conc	Dose				
Adrenaline	1:10,000	0.1ml/kg	0.3ml	0.4ml	0.5ml	0.8ml
Atropine	0.6mg/ml	0.02mg/kg	0.2ml	0.2ml	0.2ml	0.3ml
Amiodarone	150mg/3ml	5 mg/kg over 20 – 60 mins Max 300 mg	0.3ml	0.4ml	0.5ml	0.8ml
Adenosine	6mg/2ml	0.1 mg/kg Max 6 mg (1 st dose) Max 12 mg(2 nd dose)	0.1ml	0.13ml	0.16ml	0.26ml
Bicarbonate 4.2%	4.2%	1mmol/kg	6ml	8ml	10ml	16ml
Bicarbonate 8.4%	8.4%	1mmol/kg	NOT TO BE USED FOR < 1 YEAR OLD			
Calcium Gluconate	10%	0.5ml/kg	1.5ml	2.0ml	2.5ml	4ml
Dextrose	50%	1ml/kg	3ml	4ml	5ml	8ml
Magnesium Sulfate	10mmol/5ml	0.2 –0.4 mmol//kg Over 5-15 mins	0.3ml - 0.6ml	0.4ml - 0.8ml	0.5 ml - 1ml	0.8 ml -1.6ml
Uncuffed ETT diameter (mm)	4+age/4		3mm	3mm	3.5mm	4mm
Length (cm) at oral	(oral) age/2 +12	(nasal) age/2 +15	9cm	10cm	10.5cm	11cm
Defibrillation		2-4J/kg	6-12J	8-16J	10-20J	15-30J
Synchronized Cardioversion		0.5 – 2 J/kg	1.5 – 6 J	2 - 8 J	2.5-10J	4 -16J
Chest tube			8Fr	8Fr	10Fr	10Fr
Urine Catheter			6Fr	6Fr	8Fr	8Fr

QUICK REFERENCE GUIDE FOR PAEDIATRIC EMERGENCIES / CALL CODE BLUE *27

AGE			1 YR	3 YRS	5 YRS	7 YRS	10 YRS
WEIGHT			10 KG	15 KG	18 KG	22 KG	30 KG
	Conc	Dose					
Adrenaline	1:10,000	0.1ml/kg	1ml	1.5ml	1.8ml	2.2ml	3ml
Atropine	0.6mg/ml	0.02mg/kg	0.3ml	0.5ml	0.6ml	0.7ml	1ml
Amiodarone	150mg/3ml	5 mg/kg over 20 – 60 mins Max 300 mg	1ml	1.5ml	1.8ml	2.2ml	3ml
Adenosine	6mg/2ml	0.1 mg/kg Max 6 mg (1 st dose) Max 12 mg (2 nd dose)	0.3ml	0.5ml	0.6ml	0.74ml	1ml
Bicarbonate	4.2%	1mmol/kg	20ml	30ml	36ml	44ml	60ml
Bicarbonate	8.4%	1mmol/kg	10ml	15ml	18ml	22ml	30ml
Calcium Gluconate	10%	0.5ml/kg	5ml	7.5ml	9ml	11ml	15ml
Dextrose	50%	1ml/kg	10ml	15ml	18ml	22ml	30ml
Magnesium Sulfate	10mmol/5ml	0.2 –0.4 mmol//kg Over 5-15 mins	1ml - 2ml	1.5ml - 3ml	1.8ml - 3.6ml	2.2ml - 4.4ml	3ml - 6ml
Uncuffed ETT diameter (mm)	4+age/4		4mm	4.5mm	5mm	6mm	6.5mm
Length (cm) at oral	(oral) age/2 +12	(nasal) age/2 +15	12cm	13cm	14cm	16cm	17cm
Defibrillation		2-4J/kg	20-40J	30-60J	40-80J	50-90J	60-120J
Synchronized Cardioversion		0.5 – 2 J/kg	5-20J/	8–30J	9-36J/kg	11-40J	15-60J
Chest tube			10Fr	12Fr	14Fr	16Fr	16Fr
Urine Catheter			8Fr	10Fr	12Fr	14Fr	16Fr

- e. Clinic BP normative data by age and height centiles (Pediatrics 2017):
i. Boys 1-17 years old:

Age (yr)	BP percentile (+mmHg)	SBP (mmHg)							DBP (mmHg)						
		Percentile of Height							Percentile of Height						
		5 th	10 th	25 th	50 th	75 th	90 th	95 th	5 th	10 th	25 th	50 th	75 th	90 th	95 th
1	50 th	85	85	86	86	87	88	88	40	40	40	41	41	42	42
	90 th	98	99	99	100	100	101	101	52	52	53	53	54	54	54
	95 th	102	102	103	103	104	105	105	54	54	55	55	56	57	57
	95 th +12	114	114	115	115	116	117	117	66	66	67	67	68	69	69
2	50 th	87	87	88	89	89	90	91	43	43	44	44	45	46	46
	90 th	100	100	101	102	103	103	104	55	55	56	56	57	58	58
	95 th	104	105	105	106	107	107	108	57	58	58	59	60	61	61
	95 th +12	116	117	117	118	119	119	120	69	70	70	71	72	73	73
3	50 th	88	89	89	90	91	92	92	45	46	46	47	48	49	49
	90 th	101	102	102	103	104	105	105	58	58	59	59	60	61	61
	95 th	106	106	107	107	108	109	109	60	61	61	62	63	64	64
	95 th +12	118	118	119	119	120	121	121	72	73	73	74	75	76	76
4	50 th	90	90	91	92	93	94	94	48	49	49	50	51	52	52
	90 th	102	103	104	105	105	106	107	60	61	62	62	63	64	64
	95 th	107	107	108	108	109	110	110	63	64	65	66	67	67	68
	95 th +12	119	119	120	120	121	122	122	75	76	77	78	79	79	80
5	50 th	91	92	93	94	95	96	96	51	51	52	53	54	55	55
	90 th	103	104	105	106	107	108	108	63	64	65	65	66	67	67
	95 th	107	108	109	109	110	111	112	66	67	68	69	70	70	71
	95 th +12	119	120	121	121	122	123	124	78	79	80	81	82	82	83
6	50 th	93	93	94	95	96	97	98	54	54	55	56	57	57	58
	90 th	105	105	106	107	109	110	110	66	66	67	68	68	69	69
	95 th	108	109	110	111	112	113	114	69	70	70	71	72	72	73
	95 th +12	120	121	122	123	124	125	126	81	82	82	83	84	84	85
7	50 th	94	94	95	97	98	98	99	56	56	57	58	58	59	59
	90 th	106	107	108	109	110	111	111	68	68	69	70	70	71	71
	95 th	110	110	111	112	114	115	116	71	71	72	73	73	74	74
	95 th +12	122	122	123	124	126	127	128	83	83	84	85	85	86	86
8	50 th	95	96	97	98	99	99	100	57	57	58	59	59	60	60
	90 th	107	108	109	110	111	112	112	69	70	70	71	72	72	73
	95 th	111	112	112	114	115	116	117	72	73	73	74	75	75	75
	95 th +12	123	124	124	126	127	128	129	84	85	85	86	87	87	87
9	50 th	96	97	98	99	100	101	101	57	58	59	60	61	62	62
	90 th	107	108	109	110	112	113	114	70	71	72	73	74	74	74
	95 th	112	112	113	115	116	118	119	74	74	75	76	76	77	77
	95 th +12	124	124	125	127	128	130	131	86	86	87	88	88	89	89
10	50 th	97	98	99	100	101	102	103	59	60	61	62	63	63	64
	90 th	108	109	111	112	113	115	116	72	73	74	74	75	75	76
	95 th	112	113	114	116	118	120	121	76	76	77	77	78	78	78
	95 th +12	124	125	126	128	130	132	133	88	88	89	89	90	90	90

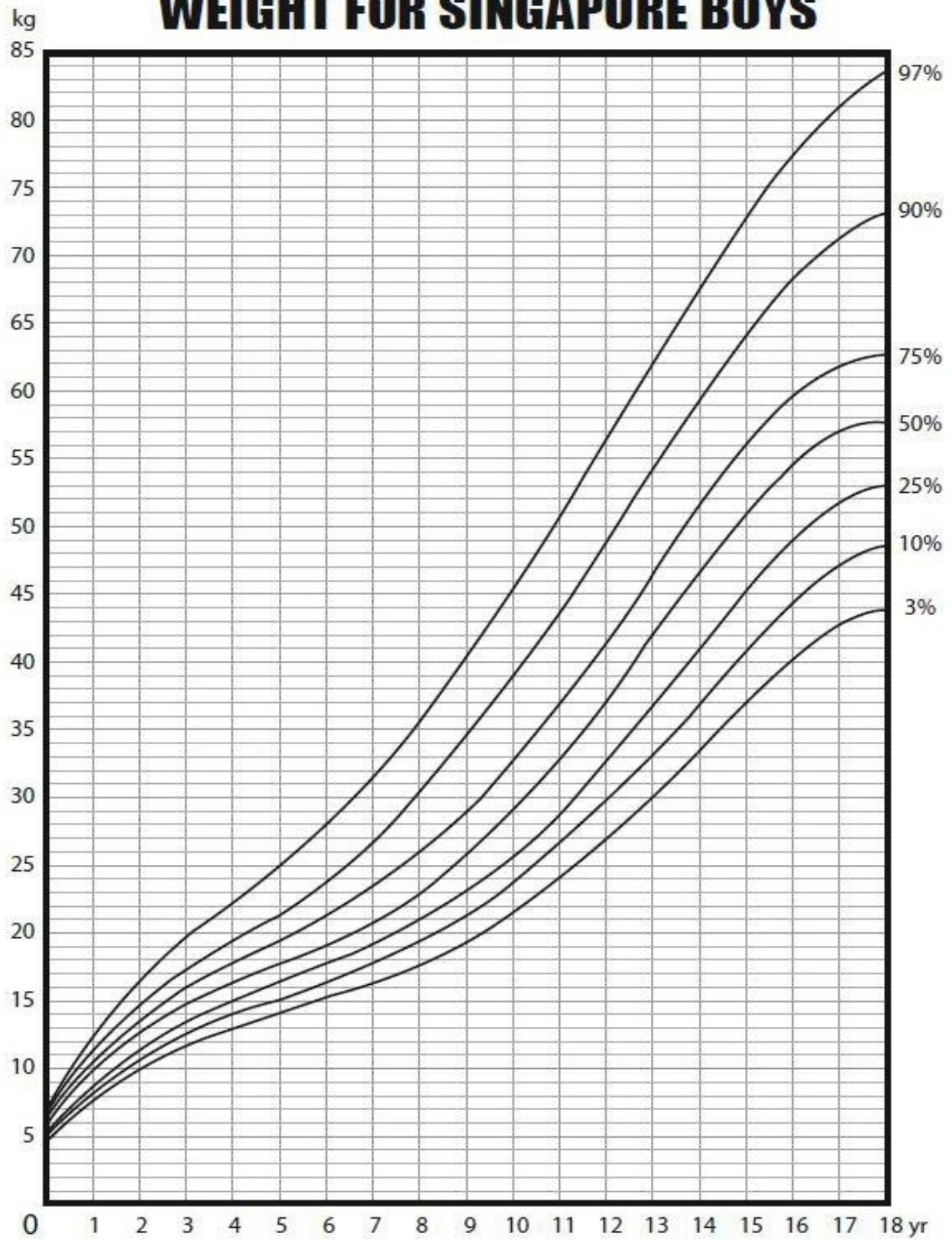
Age (yr)	BP percentile (+mmHg)	SBP (mmHg)							DBP (mmHg)						
		Percentile of Height							Percentile of Height						
		5 th	10 th	25 th	50 th	75 th	90 th	95 th	5 th	10 th	25 th	50 th	75 th	90 th	95 th
11	50 th	99	99	101	102	103	104	106	61	61	62	63	63	63	63
	90 th	110	111	112	114	116	117	118	74	74	75	75	75	76	76
	95 th	114	114	116	118	120	123	124	77	78	78	78	78	78	78
	95 th +12	126	126	128	130	132	135	136	89	90	90	90	90	90	90
12	50 th	101	101	102	104	106	108	109	61	62	62	62	62	63	63
	90 th	113	114	115	117	119	121	122	75	75	75	75	75	76	76
	95 th	116	117	118	121	124	126	128	78	78	78	78	78	79	79
	95 th +12	128	129	130	133	136	138	140	90	90	90	90	90	91	91
13	50 th	103	104	105	108	110	111	112	61	60	61	62	63	64	65
	90 th	115	116	118	121	124	126	126	74	74	74	75	76	77	77
	95 th	119	120	122	125	128	130	131	78	78	78	78	80	81	81
	95 th +12	131	132	134	137	140	142	143	90	90	90	90	92	93	93
14	50 th	105	106	109	111	112	113	113	60	60	62	64	65	66	67
	90 th	119	120	123	126	127	128	129	74	74	75	77	78	79	80
	95 th	123	125	127	130	132	133	134	77	78	79	81	82	83	84
	95 th +12	135	137	139	142	144	145	146	89	90	91	93	94	95	96
15	50 th	108	110	112	113	114	114	114	61	62	64	65	66	67	68
	90 th	123	124	126	128	129	130	130	75	76	78	79	80	81	81
	95 th	127	129	131	132	134	135	135	78	79	81	83	84	85	85
	95 th +12	139	141	143	144	146	147	147	90	91	93	95	96	97	97
16	50 th	111	112	114	115	115	116	116	63	64	66	67	68	69	69
	90 th	126	127	128	129	131	131	132	77	78	79	80	81	82	82
	95 th	130	131	133	134	135	136	137	80	81	83	84	85	86	86
	95 th +12	142	143	145	146	147	148	149	92	93	95	96	97	98	98
17	50 th	114	115	116	117	117	118	118	65	66	67	68	69	70	70
	90 th	128	129	130	131	132	133	134	78	79	80	81	82	82	83
	95 th	132	133	134	135	137	138	138	81	82	84	85	86	86	87
	95 th +12	144	145	146	147	149	150	150	93	94	96	97	98	98	99

ii. Girls 1-17 years old:

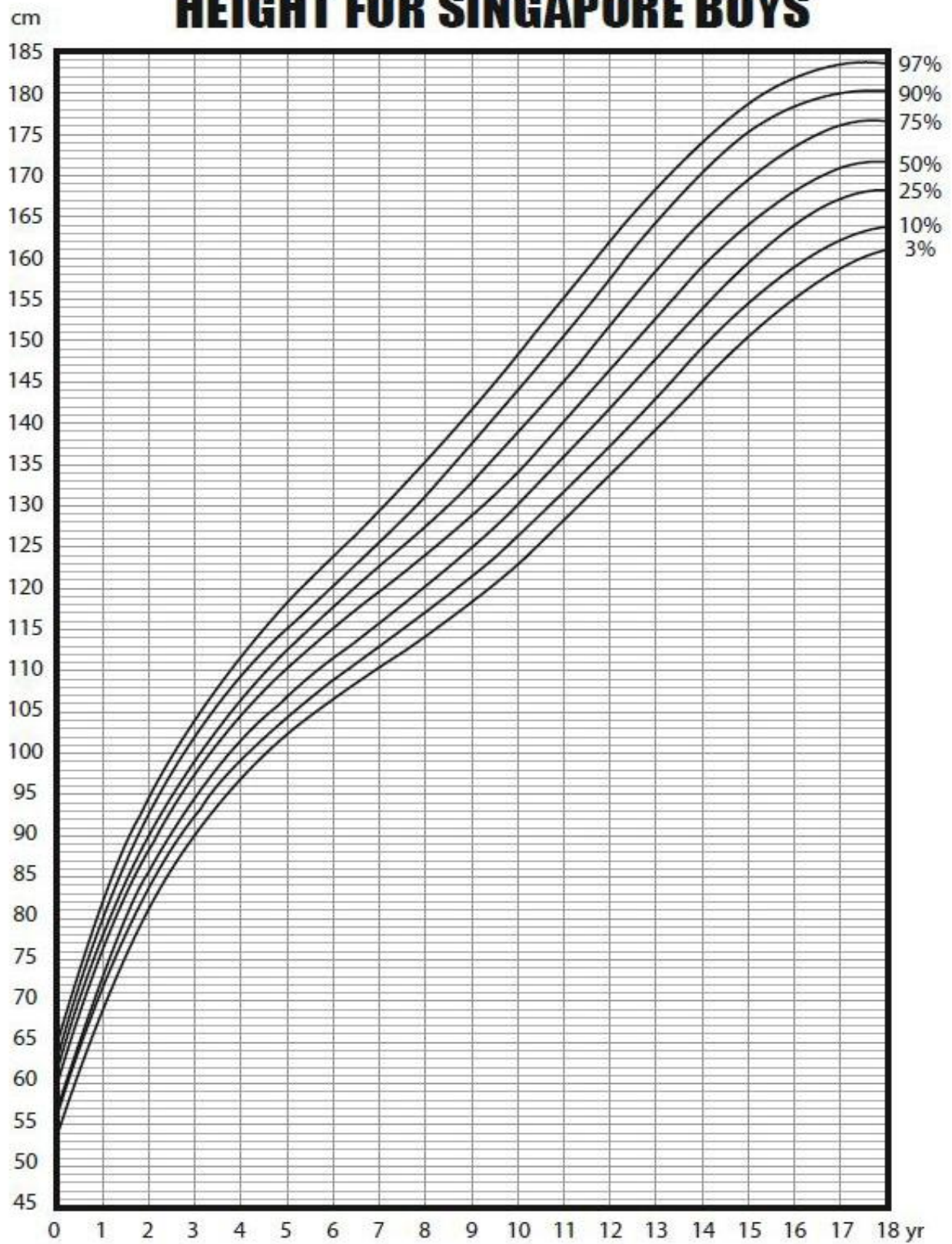
Age (yr)	BP percentile (+mmHg)	SBP (mmHg)							DBP (mmHg)						
		Percentile of Height							Percentile of Height						
		5 th	10 th	25 th	50 th	75 th	90 th	95 th	5 th	10 th	25 th	50 th	75 th	90 th	95 th
1	50 th	84	85	86	86	87	88	88	41	42	42	43	44	45	46
	90 th	98	99	99	100	101	102	102	54	55	56	56	57	58	58
	95 th	101	102	102	103	104	105	105	59	59	60	60	61	62	62
	95 th +12	113	114	114	115	116	117	117	71	71	72	72	73	74	74
2	50 th	87	87	88	89	90	91	91	45	46	47	48	49	50	51
	90 th	101	101	102	103	104	105	106	58	58	59	60	61	62	62
	95 th	104	105	106	106	107	108	109	62	63	63	64	65	66	66
	95 th +12	116	117	118	118	119	120	121	74	75	75	76	77	78	78
3	50 th	88	89	89	90	91	92	93	48	48	49	50	51	53	53
	90 th	102	103	104	104	105	106	107	60	61	61	62	63	64	65
	95 th	106	106	107	108	109	110	110	64	65	65	66	67	68	69
	95 th +12	118	118	119	120	121	122	122	76	77	77	78	79	80	81
4	50 th	89	90	91	92	93	94	94	50	51	51	53	54	55	55
	90 th	103	104	105	106	107	108	108	62	63	64	65	66	67	67
	95 th	107	108	109	109	110	111	112	66	67	68	69	70	70	71
	95 th +12	119	120	121	121	122	123	124	78	79	80	81	82	82	83
5	50 th	90	91	92	93	94	95	96	52	52	53	55	56	57	57
	90 th	104	105	106	107	108	109	110	64	65	66	67	68	69	70
	95 th	108	109	109	110	111	112	113	68	69	70	71	72	73	73
	95 th +12	120	121	121	122	123	124	125	80	81	82	83	84	85	85
6	50 th	92	92	93	94	96	97	97	54	54	55	56	57	58	59
	90 th	105	106	107	108	109	110	111	67	67	68	69	70	71	71
	95 th	109	109	110	111	112	113	114	70	71	72	72	73	74	74
	95 th +12	121	121	122	123	124	125	126	82	83	84	84	85	86	86
7	50 th	92	93	94	95	97	98	99	55	55	56	57	58	59	60
	90 th	106	106	107	109	110	111	112	68	68	69	70	71	72	72
	95 th	109	110	111	112	113	114	115	72	72	73	73	74	74	75
	95 th +12	121	122	123	124	125	126	127	84	84	85	85	86	86	87
8	50 th	93	94	95	97	98	99	100	56	56	57	59	60	61	61
	90 th	107	107	108	110	111	112	113	69	70	71	72	72	73	73
	95 th	110	111	112	113	115	116	117	72	73	74	74	75	75	75
	95 th +12	122	123	124	125	127	128	129	84	85	86	86	87	87	87
9	50 th	95	95	97	98	99	100	101	57	58	59	60	60	61	61
	90 th	108	108	109	111	112	113	114	71	71	72	73	73	73	73
	95 th	112	112	113	114	116	117	118	74	74	75	75	75	75	75
	95 th +12	124	124	125	126	128	129	130	86	86	87	87	87	87	87
10	50 th	96	97	98	99	101	102	103	58	59	59	60	61	61	62
	90 th	109	110	111	112	113	115	116	72	73	73	73	73	73	73
	95 th	113	114	114	116	117	119	120	75	75	76	76	76	76	76
	95 th +12	125	126	126	128	129	131	132	87	87	88	88	88	88	88
11	50 th	98	99	101	102	104	105	106	60	60	60	61	62	63	64
	90 th	111	112	113	114	116	118	120	74	74	74	74	74	75	75
	95 th	115	116	117	118	120	123	124	76	77	77	77	77	77	77
	95 th +12	127	128	129	130	132	135	136	88	89	89	89	89	89	89

Age (yr)	BP percentile (+mmHg)	SBP (mmHg)							DBP (mmHg)						
		Percentile of Height							Percentile of Height						
		5 th	10 th	25 th	50 th	75 th	90 th	95 th	5 th	10 th	25 th	50 th	75 th	90 th	95 th
12	50 th	102	102	104	105	107	108	108	61	61	61	62	64	65	65
	90 th	114	115	116	118	120	122	122	75	75	75	75	76	76	76
	95 th	118	119	120	122	124	125	126	78	78	78	78	79	79	79
	95 th +12	130	131	132	134	136	137	138	90	90	90	90	91	91	91
13	50 th	104	105	106	107	108	108	109	62	62	63	64	65	65	66
	90 th	116	117	119	121	122	123	123	75	75	75	76	76	76	76
	95 th	121	122	123	124	126	126	127	79	79	79	79	80	80	81
	95 th +12	133	134	135	136	138	138	139	91	91	91	91	92	92	93
14	50 th	105	106	107	108	109	109	109	63	63	64	65	66	66	66
	90 th	118	118	120	122	123	123	123	76	76	76	76	77	77	77
	95 th	123	123	124	125	126	127	127	80	80	80	80	81	81	82
	95 th +12	135	135	136	137	138	139	139	92	92	92	92	93	93	94
15	50 th	105	106	107	108	109	109	109	64	64	64	65	66	67	67
	90 th	118	119	121	122	123	123	124	76	76	76	77	77	78	78
	95 th	124	124	125	126	127	127	128	80	80	80	81	82	82	82
	95 th +12	136	136	137	138	139	139	140	92	92	92	93	94	94	94
16	50 th	106	107	108	109	109	110	110	64	64	65	66	66	67	67
	90 th	119	120	122	123	124	124	124	76	76	76	77	78	78	78
	95 th	124	125	125	127	127	128	128	80	80	80	81	82	82	82
	95 th +12	136	137	137	139	139	140	140	92	92	92	93	94	94	94
17	50 th	107	108	109	110	110	110	111	64	64	65	66	66	66	67
	90 th	120	121	123	124	124	125	125	76	76	77	77	78	78	78
	95 th	125	125	126	127	128	128	128	80	80	80	81	82	82	82
	95 th +12	137	137	138	139	140	140	140	92	92	92	93	94	94	94

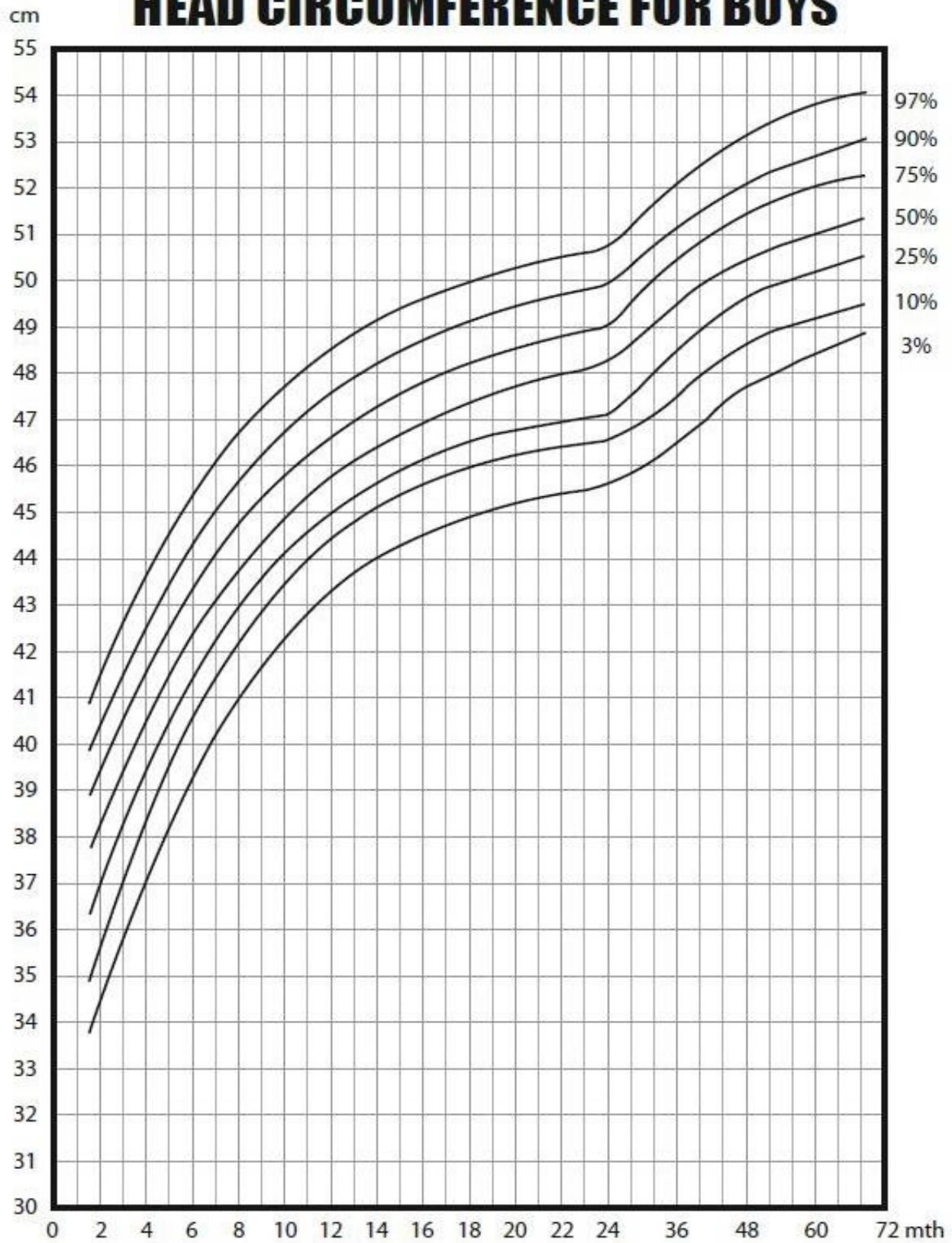
WEIGHT FOR SINGAPORE BOYS



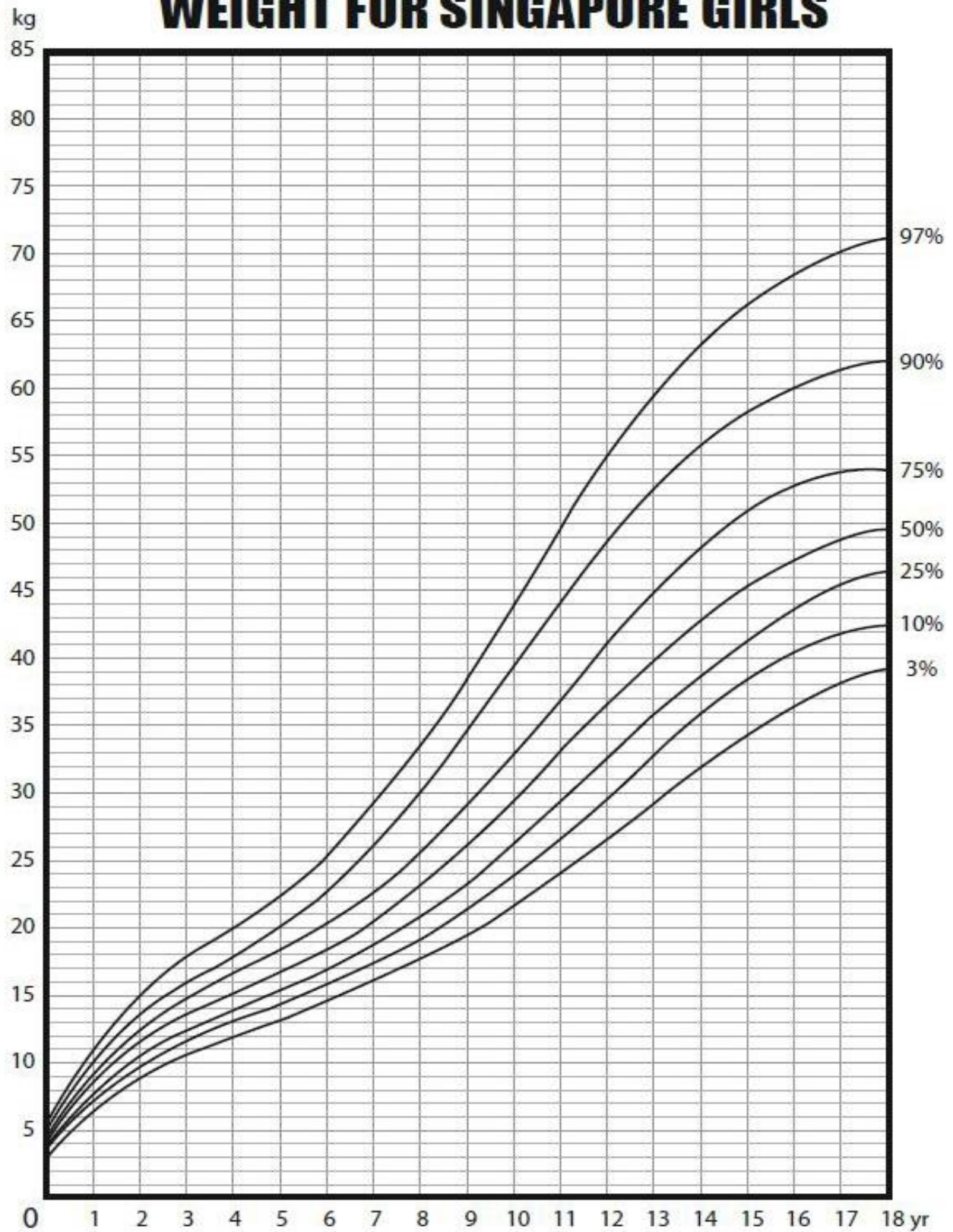
HEIGHT FOR SINGAPORE BOYS



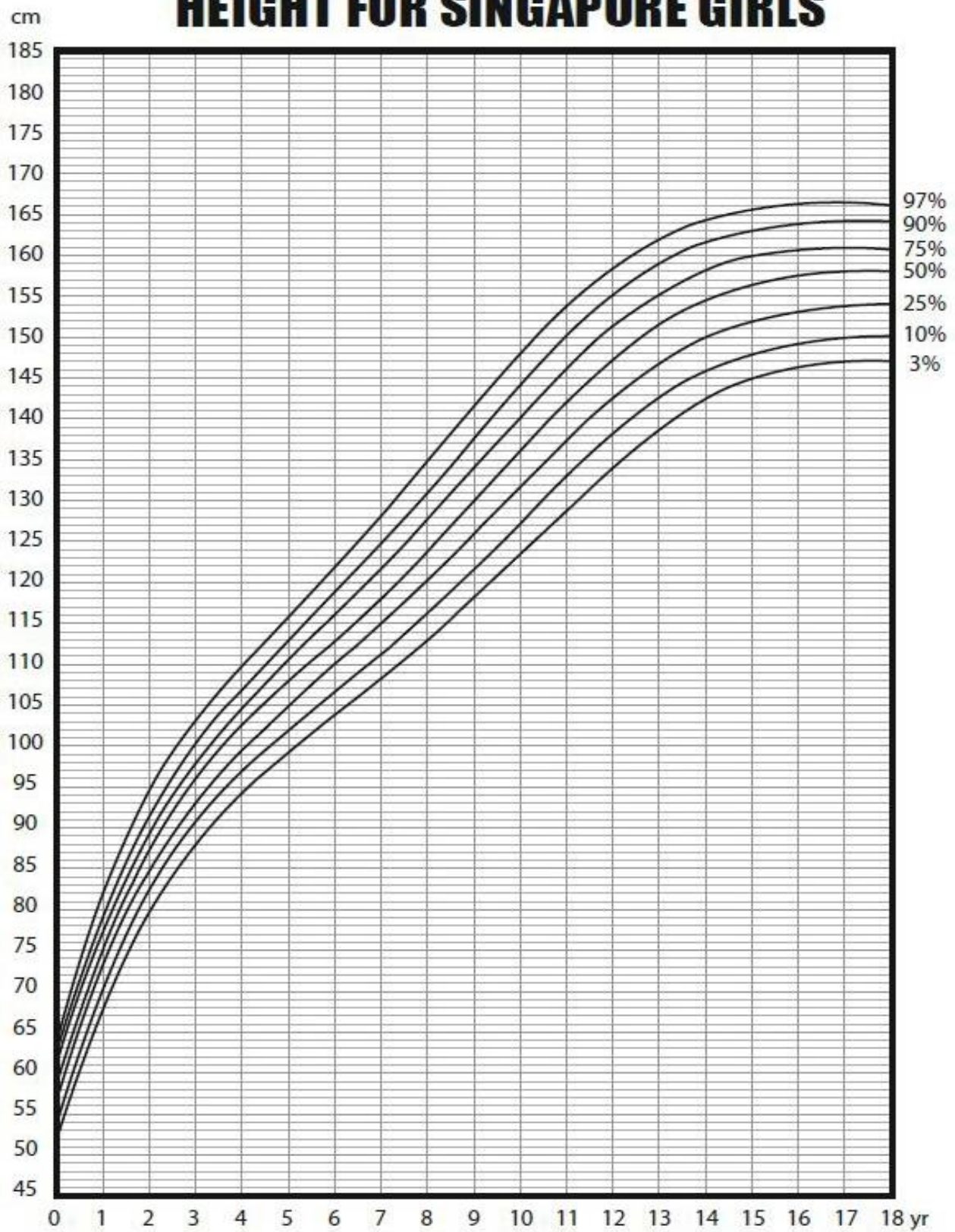
HEAD CIRCUMFERENCE FOR BOYS



WEIGHT FOR SINGAPORE GIRLS



HEIGHT FOR SINGAPORE GIRLS



HEAD CIRCUMFERENCE FOR GIRLS

